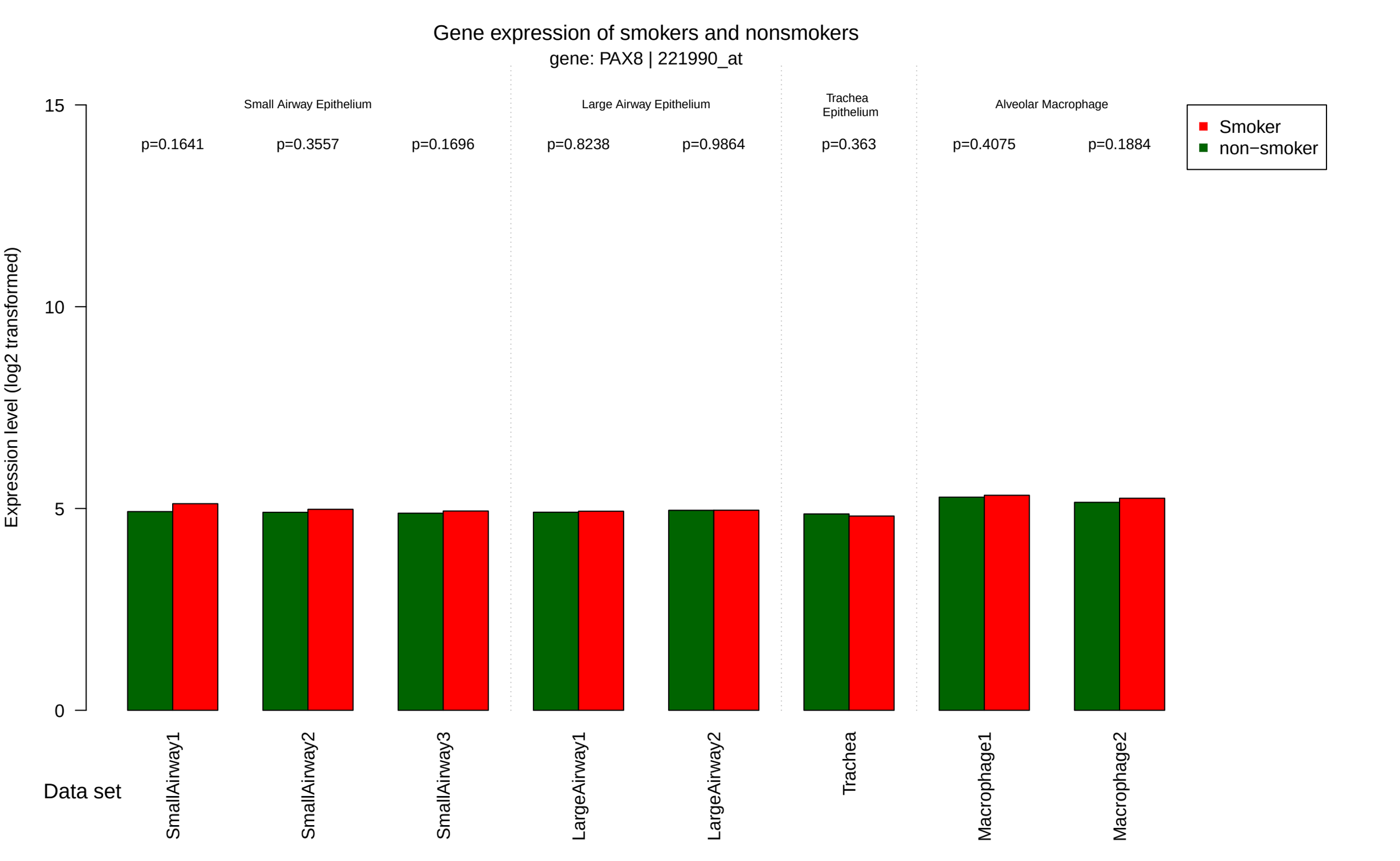
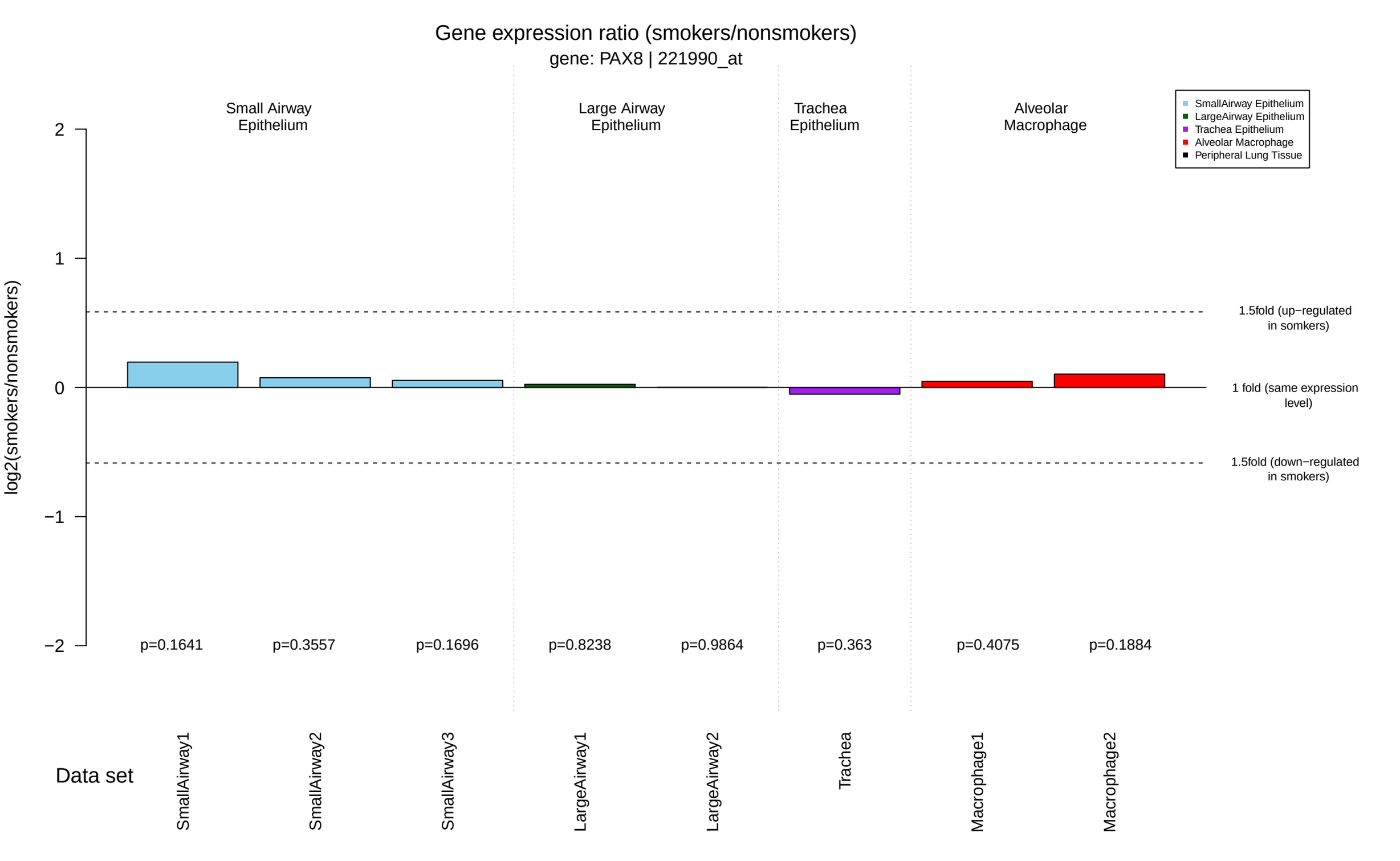
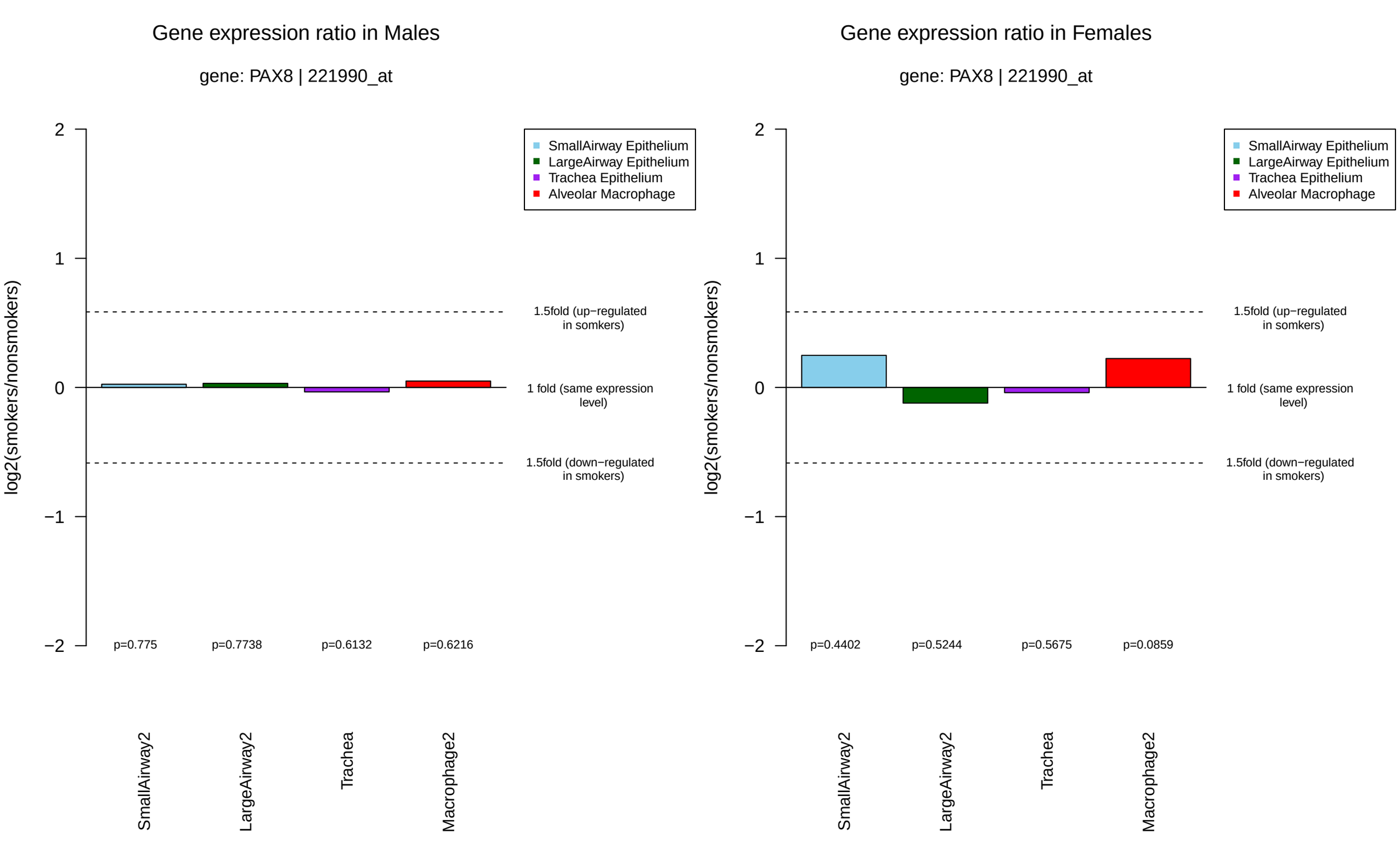
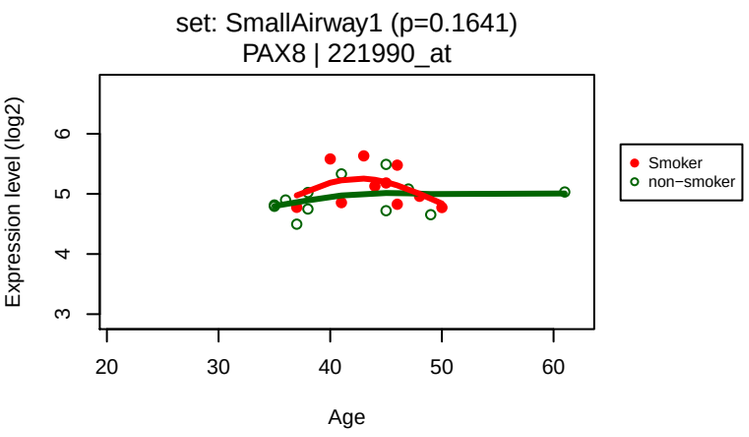
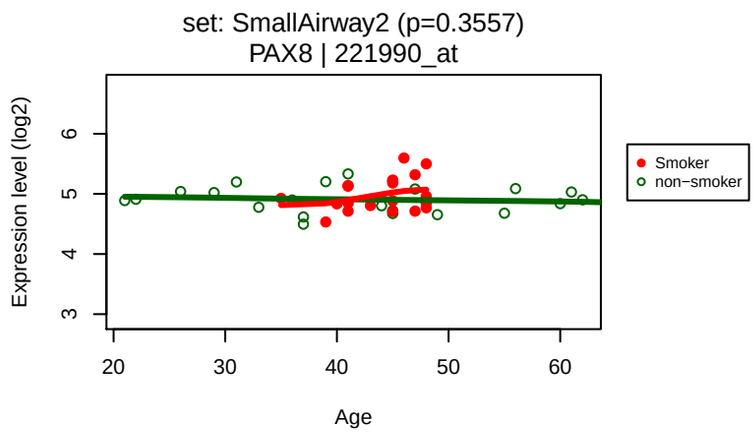
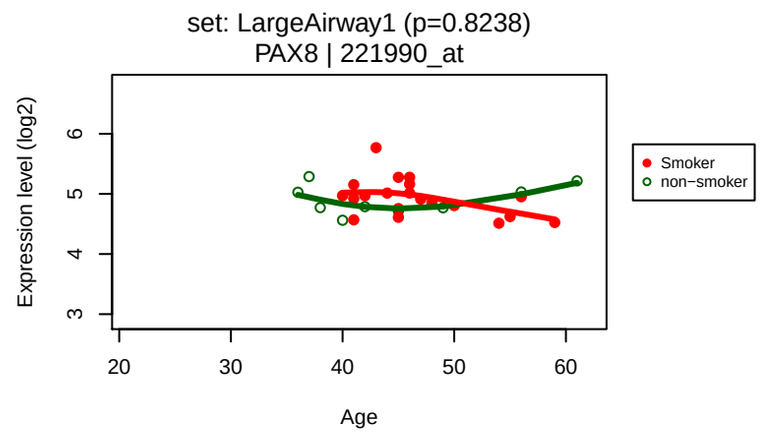

Gene expression of smokers and nonsmokers
gene: PAX8 | 221990_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1641
p=0.3557
p=0.1696
p=0.8238
p=0.9864
p=0.363
p=0.4075
p=0.1884
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

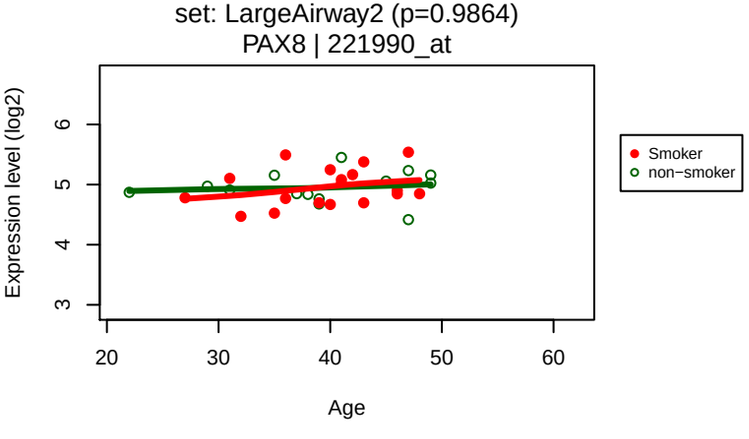

Gene expression ratio (smokers/nonsmokers)
gene: PAX8 | 221990_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1641
p=0.3557
p=0.1696
p=0.8238
p=0.9864
p=0.363
p=0.4075
p=0.1884
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

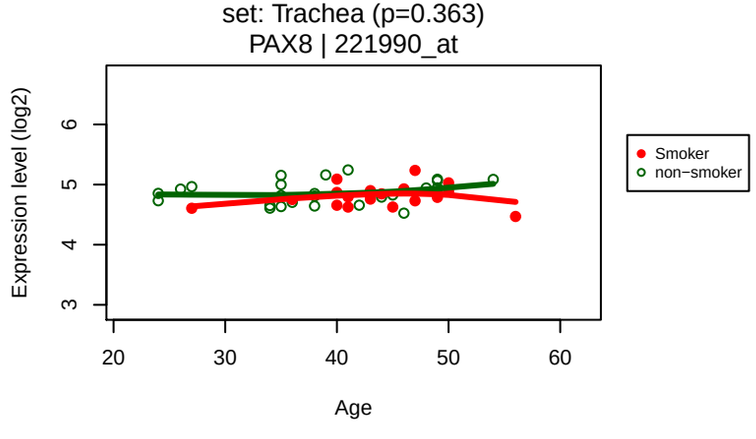

Gene expression ratio in Males
gene: PAX8 | 221990_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.775
p=0.7738
p=0.6132
p=0.6216
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAX8 | 221990_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4402
p=0.5244
p=0.5675
p=0.0859
SmallAirway2
LargeAirway2
Trachea
Macrophage2

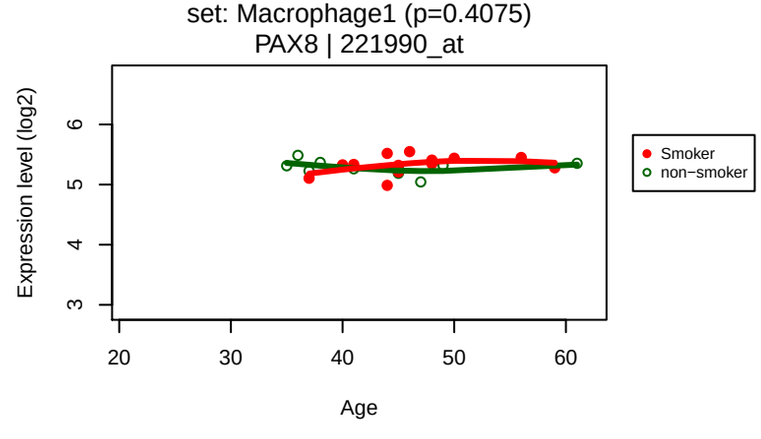

set: SmallAirway1 (p=0.1641)
PAX8 | 221990_at
Expression level (log2)
Age
Smoker
non−smoker

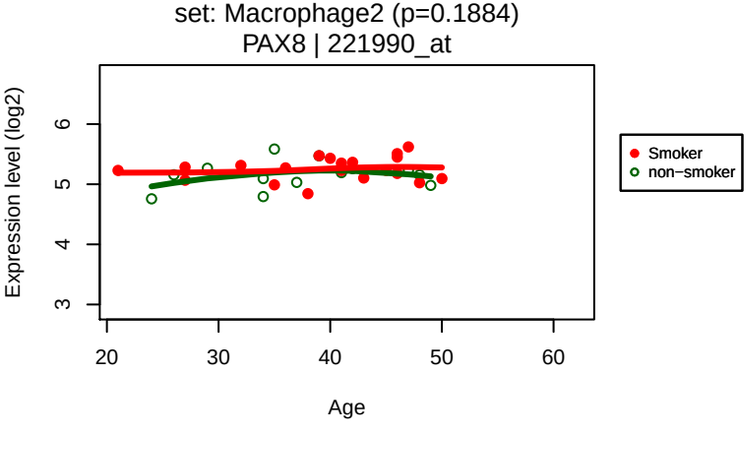

set: SmallAirway2 (p=0.3557)
PAX8 | 221990_at
Expression level (log2)
Age
Smoker
non−smoker

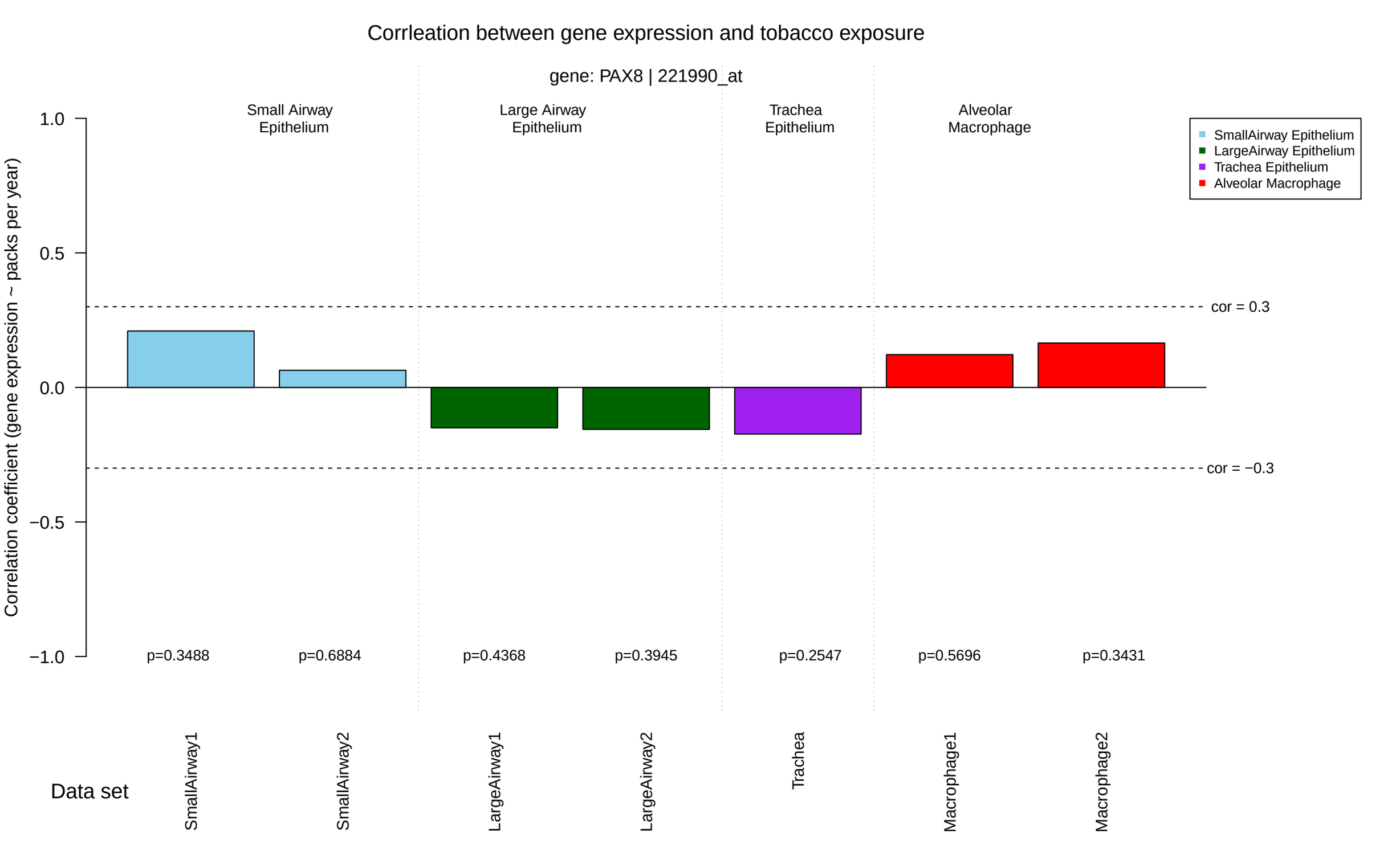

set: LargeAirway1 (p=0.8238)
PAX8 | 221990_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9864)
PAX8 | 221990_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.363)
PAX8 | 221990_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4075)
PAX8 | 221990_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1884)
PAX8 | 221990_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAX8 | 221990_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3488
p=0.6884
p=0.4368
p=0.3945
p=0.2547
p=0.5696
p=0.3431
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2